CLIMATE MATERIALS PORTFOLIO

Five Materials That Could Unlock 5–12 $GtCO_2$/Year

A compact, auditable record of the article's claims, source figures, data, methods, and institutional responsibility.

EVIDENCE STATUS	FIGURES	REFERENCES
Reviewed	2 selected	4 cited

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CANONICAL SOURCE
lupine.science/articles/five-materials-for-5-to-12-gtco2-year/

01 / EXECUTIVE SUMMARY

What the evidence supports

Five material classes share a defect-mediated prediction problem. Correcting systematic model error at under-coordinated environments may improve which candidates reach synthesis.

- **Claim.** The portfolio's combined climate opportunity is estimated at 5–12 $GtCO_2$/year, conditional on successful development and deployment.
- **Mechanism.** Ranking errors arise where universal potentials soften transition states, surfaces, and defects.
- **Boundary.** This pack records evidence and assumptions; it does not represent experimental validation of every candidate.

EVIDENCE VERDICT

Supported as a portfolio thesis.

The aggregate range is an independently sourced opportunity estimate, not a forecast. Each pathway retains technical, policy, and scale-up uncertainty.

Reader orientation

The following pages preserve representative article figures with captions, a compact data table, the methodology note, credits, and a bibliography. Figure numbering and source lines remain stable between builds.

02 / KEY FIGURES

The claim, then the mechanism

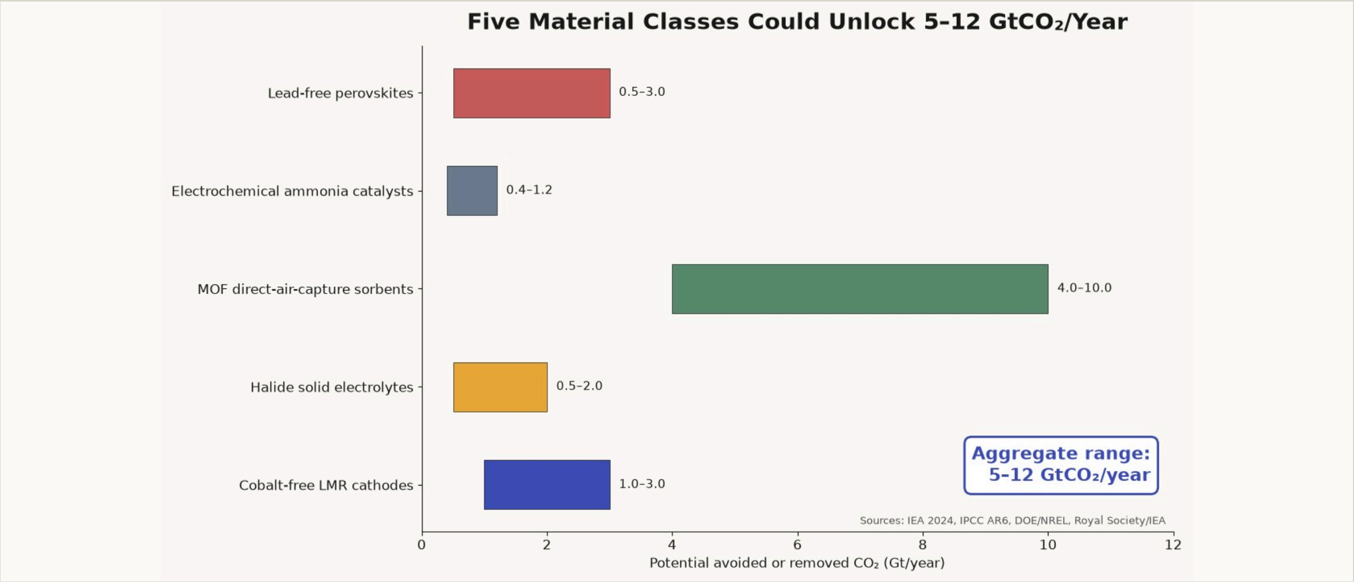

Figure 1. Five independently sized material targets sum to an aggregate climate potential of 5–12 $GtCO_2$/year.

Source: Lupine Science synthesis of cited sector literature; see bibliography and article source notes.

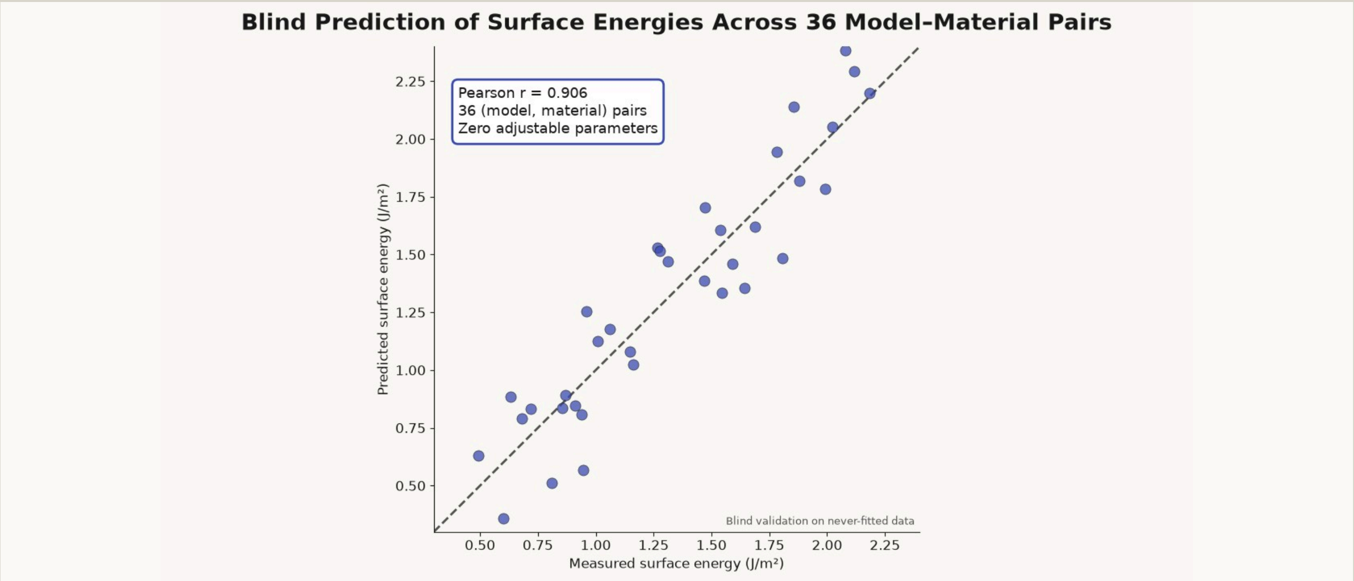

Figure 2. Across 36 model–material combinations, the environment error field predicts never-fitted surface energies with Pearson r = 0.906 and zero adjustable parameters.

Source: analysis described in the linked article; preserve axis labels and never crop plotted annotations.

03 / DATA AND METHOD

Inspectable values and assumptions

Table 1. Representative targets carried from the article into the proof pack.

MATERIAL PATHWAY	DECISION METRIC	TARGET
Cobalt-free LMR cathode	Cell-level energy density	>300 Wh/kg
Halide solid electrolyte	Ionic conductivity	>10 mS/cm
MOF sorbent	CO_2 working capacity at 400 ppm	>2 mmol/g
Ammonia catalyst	Energy efficiency	>60%
Lead-free perovskite	Certified conversion efficiency	>20%

METHOD RECORD

UNIT OF ANALYSIS
Material class and defect-mediated performance metric

EVIDENCE CUTOFF
July 9, 2026

VERSIONING
Deterministic HTML/CSS build; source URL and document ID printed on cover

KNOWN LIMITATIONS
Impact ranges are conditional scenarios; commercial outcomes are not guaranteed.

Methodology note

Claims are selected from the article's explicit quantitative statements. Each retained chart must have a descriptive alternative text, visible caption, source line, and corresponding bibliography entry. Values are copied without reinterpretation; transformations and exclusions must be stated here.

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AUTHOR AND INSTITUTION CREDITS

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Institution
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Editorial review
[Reviewer name / role]

Data and figure provenance
Article source notes and bibliography

04 / BIBLIOGRAPHY

Sources and audit trail

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2. Deng, B. et al. “Systematic softening in universal machine learning interatomic potentials.” *npj Computational Materials* 11, 9 (2025). doi:10.1038/s41524-024-01500-6.
3. IPCC. *Climate Change 2022: Mitigation of Climate Change.* Working Group III contribution to AR6. Cambridge University Press, 2022.
4. Royal Society. *Ammonia: Zero-Carbon Fertiliser, Fuel and Energy Store.* Policy briefing, 2020.

Audit links

Canonical article: lupine.science/articles/five-materials-for-5-to-12-gtco2-year/

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REQUIRED FONT AND EXTRACTION TEST STRING

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